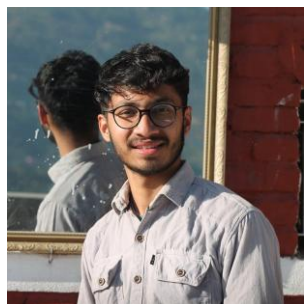


DOCSGUIDE

An Assistant for Government Document Procedures

Team Members



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Data Ingestion &
Embeddings



Bikash Pokhrel

Retriever Design &
RAG Evaluation



Dipin Adhikari

Web App &
Retriever Design



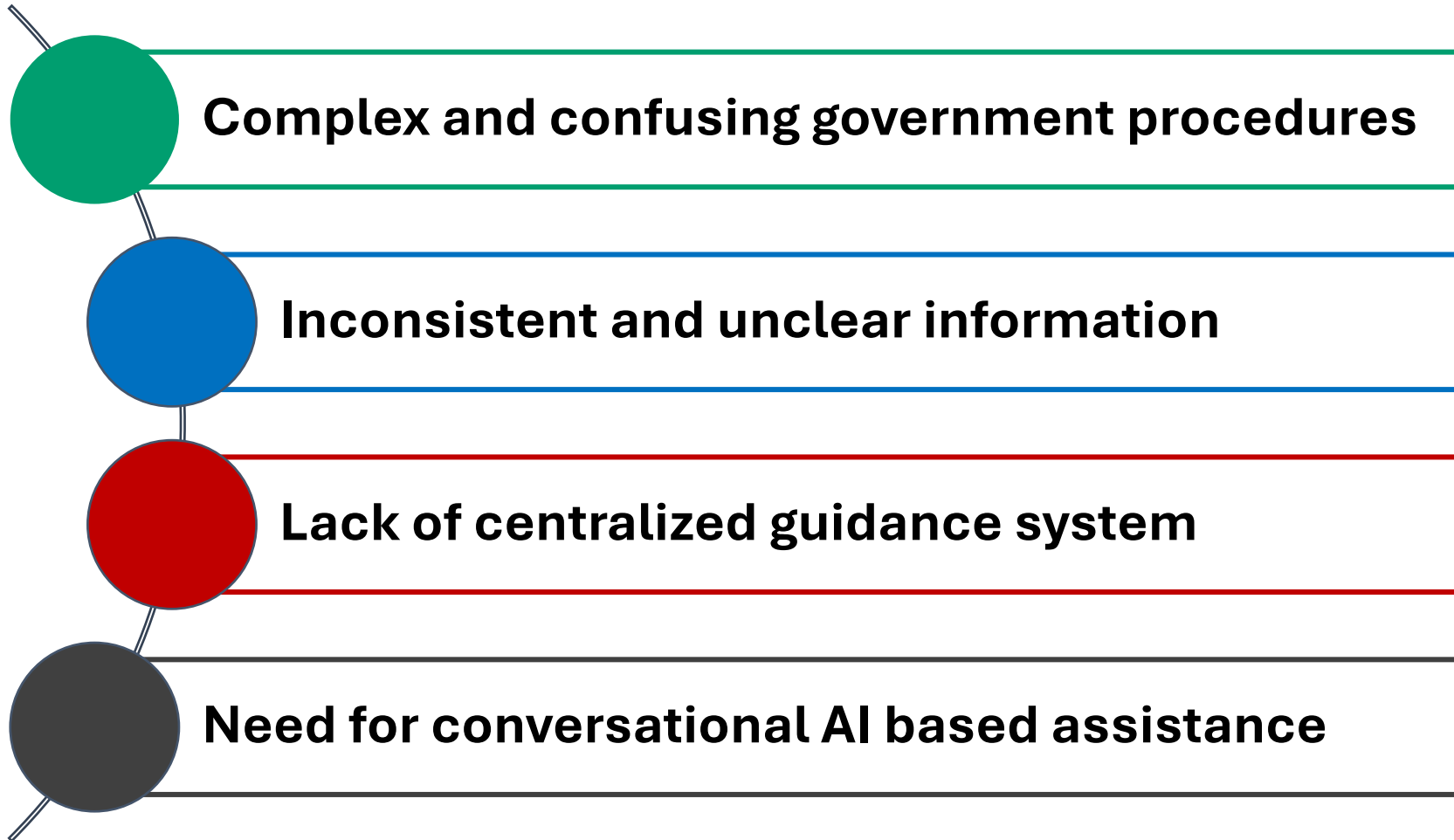
Utsab Dahal

Generator Design &
RAG Evaluation

Project Overview

- **DocsGuide** is an AI chatbot that helps Nepali citizens in understanding government document procedures such as obtaining citizenship certificates, passports etc.
- It uses a **Retrieval-Augmented Generation (RAG)** model to deliver accurate, step-by-step guidance based on verified official sources.
- The system supports **both Nepali and English**, ensuring accessibility for diverse users.
- Users can **ask questions or seek legal information** related to government documents.

Problem Statement



Objectives

- 1 To develop an AI chatbot that guides users through government document procedures.
- 2 To provide accurate, verified, and bilingual information for document related queries.
- 3 Evaluate system performance based on accuracy and response quality.

Methodology

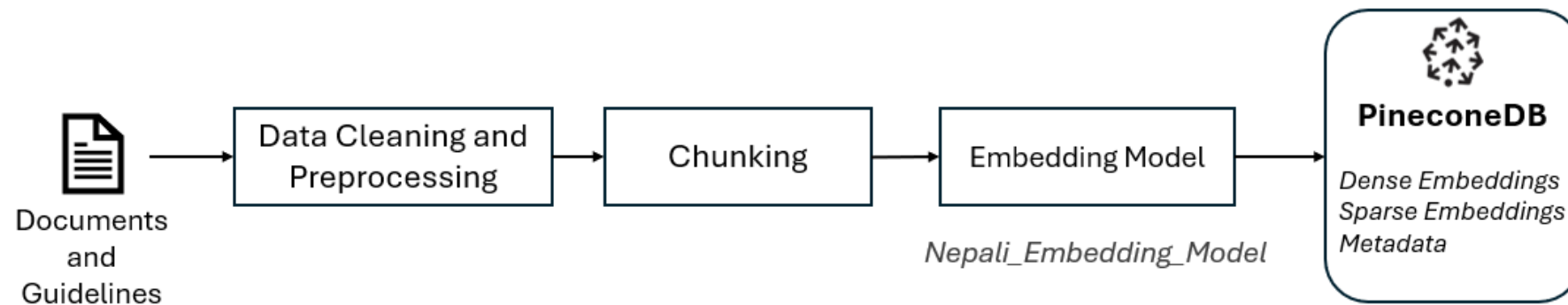
1. Document Collection and Chunking

- Collected and preprocessed verified government documents related to citizenship and passport procedures (acts, rules, FAQs, guidelines).
- Applied chunking strategies for efficient retrieval:
 - Section-based chunking*** for structured division.
 - Metadata assignment*** — document type, references, section/subsection, tags (question category), and source links.

Methodology

2. Embedding Generation and Storage

- Embedding Model *universalml/Nepali_Embedding_Model* with:
Maximum sequence length: 512 tokens
Output dimensionality: 1024
- Generated & Stored dense embeddings with metadata in *PineconeDB*.
- Sparse vector representations using BM25 encoder to support keyword-based search.



Methodology

3. Retriever Design

Multiple retrieval strategies:

- **Pure Dense Search** — similarity search using dense embeddings only.
- **Hybrid Search** — combination of dense embeddings and keyword-based BM25 search(Controlled by alpha parameter)
- **Metadata-based Filtering** — filtered chunks by document type, tags.
- **Cross-Encoder Reranking** — refined top results using cross encoder model (jinaai/jina-reranker-v2-base-multilingual) for higher relevance.

Query Translation and Rewriting Layer — translates queries to Nepali, includes chat context, and detects language.

Methodology

4. Generator Design

- **Language-Aware Response Generation** : Gemini API to generate answers with given context in the detected language.
- **Source Citation** : Cite references for retrieved information.

5. Web App Design

- **Frontend**: Next.js & Tailwind CSS.
- **Backend**: FastAPI

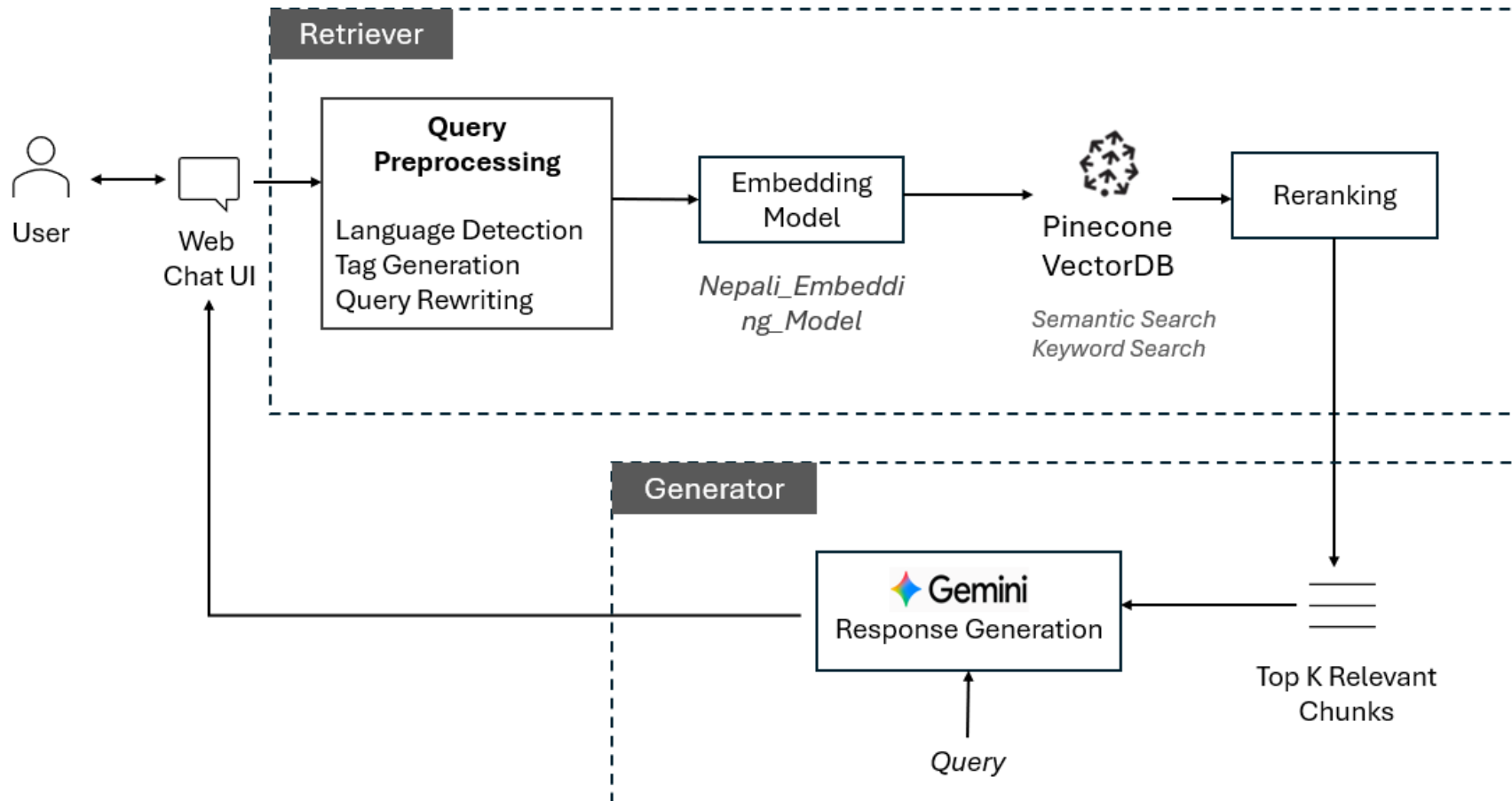
Methodology

6. Evaluation and Testing

- **Evaluation Dataset:** 28 question & answer pairs.
- **Testing Dataset:** 15 question & answer pairs.
- Manually labelling of relevant document chunks for each question to serve as ground truth.
- RAG evaluation pipeline to calculate metrics :
 - Recall** – proportion of relevant chunks retrieved
 - Precision** – proportion of retrieved chunks that were relevant,
 - Answer Correctness** – accuracy of generated responses against labeled data.

System Design

System Architecture



Results

Evaluation Configuration	Metrics
Without Metadata Filtering and without re-ranking	
• Pure Dense Search • Top 3 used for Generation	Answer Correctness: 78.92% Recall@7 : 89.94% Precision@7: 21.93%
• Hybrid Search • 50% Dense and 50% Sparse • Top 3 used for Generation	Answer Correctness: 83.14% Recall@7 : 90.23% Precision@7: 21.93%
• Hybrid Search • 75% Dense and 25% Sparse • Top 3 used for Generation	Answer Correctness: 84.10% Recall@7 : 90.23% Precision@7: 21.93%

Results

Evaluation Configuration	Metrics
Without Metadata Filtering and with re-ranking	
- Hybrid Search (50% Dense and 50% Sparse) <i>Document type only meta data filter</i> - Top 3 used for Generation	Answer Correctness: 85.77% Recall@7 : 90.23% Precision@7: 21.93%
With Metadata Filtering and with re-ranking	
- Hybrid Search (50% Dense and 50% Sparse) <i>Document Type and Tag based filter</i> <i>Top 3 used for generation</i>	Answer Correctness: 82.52% Recall@7 : 75.7% Precision@7: 19.2%
Testing Dataset	
- Hybrid Search (50% Dense and 50% Sparse) <i>Document type only meta data filter</i> - Top 3 used for Generation	Answer Correctness: 89.3% Recall@7 : 88.9% Precision@7: 21.0%

Conclusion

- Successfully developed a bilingual RAG-based chatbot to assist users with Nepali government document procedures.
- Ensured accurate and verified responses using official sources and hybrid retrieval strategies.
- **Hybrid search (50% vector + 50% keyword, with reranking)** achieved around 90% recall and 85% answer correctness.

Limitations

- Limited to citizenship and passport only
- Small evaluation dataset may not fully represent real user queries.
- Lack of large-scale real user testing.
- Semantic search limited to Nepali text, as the knowledge base was primarily in Nepali.
- Translation inaccuracies.
- Incorrect question tagging by Gemini when using tag-based filtering.

Challenges

- Selection of chunking strategies based on document type and structure.
- Balancing between keyword and semantic search to achieve optimal hybrid retrieval performance.
- Semantic accuracy of embedding and re-ranking models

Future Works

- Expand to more government documents.
- Improve legal translation with external knowledge base and mixed-language handling.
- Handle both English and Nepali knowledge bases for improved retrieval.

THANK YOU